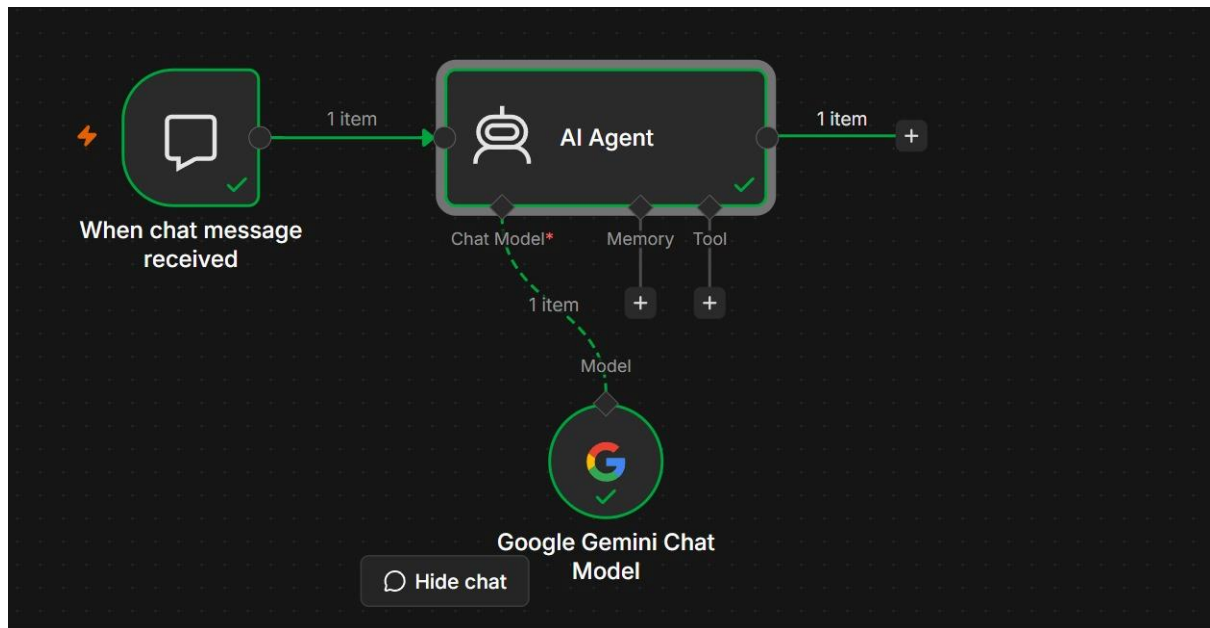


PROJECT-25

AI Resume Reviewer



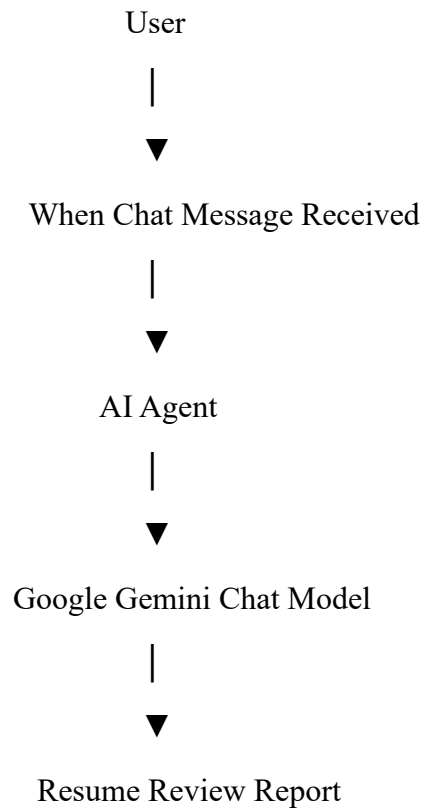
Project Objective

The objective of this project is to build an **AI-powered Resume Reviewer** that analyzes a user's resume and provides detailed feedback to improve its quality, readability, professionalism, and ATS (Applicant Tracking System) compatibility. The AI Agent identifies strengths, weaknesses, missing information, formatting issues, and offers actionable suggestions to help users create a stronger resume.

The AI Agent can review:

- Professional resumes
 - Student resumes
 - Fresher resumes
 - Experienced professional resumes
 - Technical resumes
 - Management resumes
 - ATS compatibility
 - Resume formatting
 - Skills and achievements
 - Career objective and summary
-

Workflow Architecture



(Use the same workflow image as the reference document.)

Workflow Components

Step	Node	Purpose
1	When Chat Message Received	Receives the user's resume and review request
2	AI Agent	Analyzes the resume and prepares the review prompt
3	Google Gemini Chat Model	Reviews the resume and provides suggestions

Prerequisites

Before building this project, ensure you have:

- n8n (Cloud or Self-hosted)
- Google AI Studio API Key
- Google Gemini Chat Model configured
- Internet connection

Step 1 – Create Workflow

Open n8n

Click

New Workflow

Rename it

AI Resume Reviewer

Step 2 – Add Chat Trigger

Add

When Chat Message Received

Purpose

Receives the user's resume and review request.

Example Input

Review my resume for a Software Developer position and suggest improvements to make it ATS-friendly.

Output

```
{  
  "chatInput": "Review my resume for a Software Developer position and suggest  
improvements to make it ATS-friendly."  
}
```

Step 3 – Add AI Agent

Connect

When Chat Message Received

|



AI Agent

Configure

Source for Prompt

Connected Chat Trigger Node

Prompt

```
{{ $json.chatInput }}
```

Step 4 – Configure the System Prompt

Paste the following into the **System Message**.

You are an AI Resume Reviewer.

Your task is to analyze resumes and provide detailed, professional, and constructive feedback.

Instructions:

- Carefully review the user's resume.
- Evaluate the resume's overall structure and formatting.
- Check the professional summary or career objective.
- Analyze the work experience section for clarity and impact.
- Review technical and soft skills.
- Identify missing or weak sections.
- Suggest stronger action verbs and measurable achievements.
- Evaluate ATS (Applicant Tracking System) compatibility.
- Recommend improvements to keywords based on the target job role.
- Highlight grammar, spelling, and formatting issues.
- Organize the feedback into the following sections:
 - Overall Score (out of 10)
 - Strengths
 - Areas for Improvement
 - ATS Compatibility
 - Suggested Resume Improvements
- Keep the feedback professional, actionable, and encouraging.

- Return only the resume review.

Step 5 – Add Google Gemini Chat Model

Click

+ **Chat Model**

Choose

Google Gemini Chat Model

Configuration

Setting	Value
Provider	Google Gemini
Model	Gemini 2.5 Flash (or another available Gemini model)
Authentication	Google AI Studio API Key

Connect it to the **AI Agent**.

Final Workflow

When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

Workflow Execution

Step 1

User enters

Review my resume for a Data Analyst position and suggest improvements.

↓

Step 2

The **When Chat Message Received** node receives the user's request.



Step 3

The **AI Agent** analyzes the resume, target job role, and review requirements.



Step 4

The prompt is sent to the **Google Gemini Chat Model**.



Step 5

Gemini reviews the resume, evaluates ATS compatibility, identifies strengths and weaknesses, and generates improvement suggestions.



Step 6

The completed resume review report is returned to the user.

Sample Inputs and Outputs

Example 1

Input

Review my Software Developer resume.

Output

Overall Score: 8.5/10

Strengths

- Well-organized layout.
- Strong technical skills section.
- Relevant project experience.

Areas for Improvement

- Add measurable achievements.
- Improve the professional summary.

- Include GitHub and LinkedIn profile links.

ATS Compatibility

Good, but more job-specific keywords can improve ranking.

Example 2

Input

Review my resume for a Data Analyst role.

Output

Overall Score: 9/10

Strengths

- Excellent project descriptions.
- Strong analytical skills.
- Good education section.

Areas for Improvement

- Add certifications.
 - Quantify project outcomes.
 - Improve keyword optimization.
-

Example 3

Input

Review my fresher resume.

Output

Overall Score: 7.8/10

Strengths

- Clean formatting.
- Good academic achievements.

Areas for Improvement

- Add internships.
- Include personal projects.

- Expand technical skills section.

Example 4

Input

Review my resume for an HR Manager position.

Output

Overall Score: 8.8/10

Strengths

- Strong leadership experience.
- Clear career progression.
- Professional formatting.

Areas for Improvement

- Add measurable recruitment achievements.
- Improve ATS keywords.
- Include HR software expertise.

Workflow Diagram

User

|



When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

|



Resume Review Report

Error Handling

Error: No Prompt Specified

Cause

The AI Agent did not receive the user's request.

Solution

Verify that the **Prompt** field contains:

```
{{ $json.chatInput }}
```

Error: Authentication Failed

Cause

Invalid Google Gemini API Key.

Solution

Reconnect or update your Google AI Studio credentials.

Error: Too Many Requests

Cause

Gemini API rate limit exceeded.

Solution

- Wait a few minutes.
 - Check your API quota.
 - Retry the request.
-

Learning Outcomes

After completing this project, you will be able to:

- Build an AI-powered Resume Reviewer in n8n.
- Configure an AI Agent with Google Gemini.
- Design prompts for resume analysis.

- Evaluate ATS compatibility using AI.
 - Build conversational AI workflows for career assistance.
-

Real-World Applications

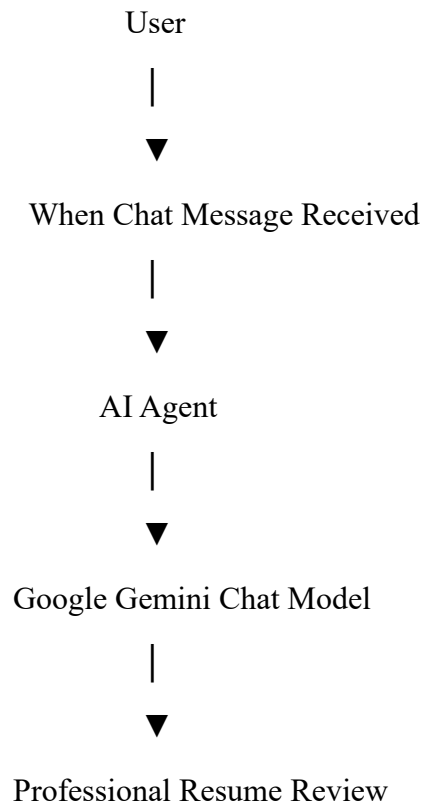
- Resume evaluation
 - Career counseling
 - ATS optimization
 - Job application assistance
 - University placement support
 - HR recruitment screening
 - Career coaching
 - Professional profile improvement
 - Interview preparation
 - Recruitment automation
-

Practice Challenge

Enhance this project by adding:

1. A target job role selector (Software Engineer, Data Analyst, HR Manager, etc.).
 2. An ATS score calculator.
 3. Resume keyword optimization based on job descriptions.
 4. A Memory node to remember the user's preferred job role.
 5. A Google Sheets node to store resume review reports.
 6. An Email node to automatically send the reviewed resume and feedback report to the user.
-

Final Workflow



Expected Result

The **AI Resume Reviewer** accepts a user's resume and target job role, uses the **Google Gemini Chat Model** to evaluate the resume's structure, content, ATS compatibility, skills, and overall effectiveness, and returns a comprehensive review report with strengths, weaknesses, improvement suggestions, and an overall score. This project teaches prompt engineering, AI-powered document analysis, and conversational workflow design in n8n, making it an excellent foundation for building intelligent career guidance and recruitment automation solutions.